MOBINA SAEDI

Computer Engineering Undergraduate | Aspiring AI/ML Researcher

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EDUCATION

B.Sc. in Computer Engineering – Islamic Azad University, Central Tehran Branch Expected Graduation: June 2027

GPA: 18/20

Selected coursework (grades out of 20)

Course	Grade	What it covered / what I did
Software Project	**20/20**	Defended my research manuscript as the project: a paired protocol testing how Persian orthographic variants (Arabic-Persian yeh and kaf) and input normalization affect LLM multiple-choice question answering, with a Python evaluation pipeline and an exploratory pilot. The manuscript is in preparation.
Internship Project	**20/20**	Built Air Drawing, a real-time computer-vision application (Python, OpenCV, MediaPipe) that tracks the hand through a webcam and turns finger gestures into drawing on a virtual canvas.
Theory of Languages and Compilers	**20/20**	Formal languages and automata, lexical analysis, parsing, and the phases of compiler construction.
Algorithms	**17/20**	Design and analysis of algorithms: complexity, divide and conquer, greedy methods, dynamic programming, and graph algorithms.

RESEARCH INTERESTS

Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Explainable AI (XAI), Low-Resource and Multilingual NLP, Data Science

RESEARCH EXPERIENCE

Persian NLP and Large Language Model Robustness Research Manuscript in Preparation – Oct 2026

Orthographic Variation and Input Normalization in Persian LLM Multiple-Choice Question Answering: A Paired Pilot Protocol

Co-author: Mohammad Farshad Fazeli Sabzevar

- Studies whether Persian orthographic variants (e.g., Arabic and Persian forms of yeh and kaf) change LLM answers on multiple-choice questions, and whether input normalization reduces the effect, using four paired conditions: reference, normalized, perturbed, and normalized-perturbed.
- Contributed to writing the research manuscript.
- Designed the paired experimental protocol for controlled orthographic perturbations and normalization.
- Implemented the Python code supporting the research workflow, and ran an exploratory pilot with Qwen2.5-1.5B-Instruct through local inference with llama.cpp. The pilot is too small for conclusions about robustness; the manuscript states this limitation.

Status: in preparation; not yet published or submitted.

PROJECTS

Air Drawing – Gesture-Based Virtual Drawing System (code) Python, OpenCV, MediaPipe

- Real-time touchless drawing: webcam hand tracking with finger key points converts gestures into drawing actions on a virtual canvas (image processing and computer vision).

EduPredict – Explainable ML for Early Identification of Students at Academic Risk (code) Python, Scikit-learn

- In progress. Pipeline for predicting academic risk from a dataset of 4,424 student records with 37 variables: exploratory analysis, preprocessing, feature engineering, and model comparison. Planned: interpretability analysis (feature importance, SHAP).

Other projects: House Price Prediction (ML regression on property features); Inventory Management System (Python); Age Calculator (code).

EXPERIENCE

AI/ML Intern – AIO Learn Oct 2026 – Present

- Building foundational machine learning knowledge through structured learning and practical training.

English Language Instructor and Online English Tutor (including freelance tutoring) Jun 2024 – Present

Safir Gofteman Institute, Melal Language Institute, AmazingTalker

- Teaching English to learners of different ages and proficiency levels, in institutes and online.

TECHNICAL SKILLS

Programming: Python
Data analysis: NumPy, Pandas
Visualization: Matplotlib, Seaborn
Machine learning: Scikit-learn; foundational knowledge of machine learning concepts
Computer vision: OpenCV, MediaPipe (Air Drawing project)
Tools: Jupyter Notebook, Git/GitHub, llama.cpp (local LLM inference)

CERTIFICATIONS

Exploratory Data Analysis for Machine Learning – IBM, Coursera Issued August 2026
Verify: coursera.org/verify/D792BPCQ0W14

Programming with Python – Mojtama Fanni (Technical and Vocational Training Complex) Issued 2024

IBM Machine Learning Professional Certificate In progress

LANGUAGES

Persian: Native
English: Advanced (C1, self-assessed). IELTS preparation in progress; target test date Spring 2027.
French: Upper-intermediate (B2, self-assessed). TCF preparation in progress; target test date Spring 2027.

I have teaching and freelance experience as an English instructor. I have not yet taken IELTS because sanctions on Iran restrict access to international testing; I am now preparing for IELTS and for the TCF.